SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 35%

QUESTION: 1 **Total Marks:100**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

Scenario: A university must schedule final examinations for 5 courses — Artificial Intelligence (A), Database Systems (B), Computer Networks (C), Operating Systems (D), and Software Engineering (E) — across 3 available exam slots: Slot1, Slot2, Slot3 (one exam per day, over three days).

Due to overlapping student enrollments, the following pairs of courses cannot be scheduled in the same slot (a shared group of students is registered in both courses):

Conflicting Pairs (must get different slots)
A – B
A – C
B – C
B – D
C – D
C – E
D – E

An intelligent scheduling agent must produce a conflict-free exam timetable satisfying all the above constraints.

Answer the following:

a) Give the PEAS description (Performance measure, Environment, Actuators, Sensors) for this exam-scheduling agent.

b) Identify and justify the type of intelligent agent (simple reflex, model-based, goal-based, or utility-based) best suited to this scheduling task.

c) Formally model this problem as a Constraint Satisfaction Problem (CSP) by defining: Variables, Domains, and Constraints. Also represent the problem as a constraint graph (list the vertices and edges).

d) Apply plain Backtracking Search to solve this CSP. Use the static variable order A → B → C → D → E and try domain values in the order Slot1 → Slot2 → Slot3. Show every assignment attempt, including failed attempts and backtracking steps, in a step-by-step table, and state the final valid timetable obtained.

e) Apply Backtracking Search with Forward Checking to the same CSP, using the same variable and value order. Show how the domains of unassigned variables are pruned after each assignment, and state the final valid timetable obtained.

f) Compare plain Backtracking Search and Backtracking with Forward Checking for this problem in terms of number of assignment attempts, number of backtracks, and overall efficiency. Explain *why* forward checking reduces the search effort.

Marks Distribution Summary:

Part	**Concept Assessed**	**Marks**
a	PEAS / Agent Modeling	10
b	Agent Type Classification	10
c	CSP Formulation + Constraint Graph	15
d	Plain Backtracking Search	25
e	Backtracking + Forward Checking	20
f	Comparative Analysis	20
Total		**100**

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided